

GENETIC TRAITS AND CROP PERFORMANCE DBMS

FIRST NORMALIZATION TABLES (1NF)

Table 1: Users

user_id	username	email	role	created_at
1	admin_user	admin@lab.edu	Admin	2024-01-15
2	researcher_1	researcher@lab.edu	Researcher	2024-01-20
3	guest_user	guest@lab.edu	Guest	2024-02-01

1NF Compliance: - ✓ **Atomic Values:** Each field contains single indivisible values (username, email, role are not composite) - ✓ **No Repeating Groups:** Each user has only one role; no multi-valued attributes like roles[] array - ✓ **Unique Identification:** user_id uniquely identifies each row, ensuring no duplicate user records

Table 2: Crop_Info

crop_id	crop_name	variety	origin_region	description
1	Wheat	Winter Wheat	North America	Hardy winter grain crop
2	Rice	Basmati	South Asia	Aromatic long-grain rice
3	Corn	Hybrid Yellow	Americas	High-yield maize variety

1NF Compliance: - ✓ **Atomic Values:** Single crop_name, variety, and origin_region per row (no composite values) - ✓ **No Repeating Groups:** One variety per crop record (not multiple varieties like variety1, variety2) - ✓ **Unique Identification:** crop_id uniquely identifies each crop variety

Table 3: Genetic_Traits

trait_id	crop_id	gene_code	category	trait_name	description
1	1	SNP_WH_001	Yield	High Yield Gene	Increases grain production
2	2	SNP_RI_002	Disease Resistance	Blast Resistance	Resistance to blast fungus
3	3	SNP_CO_003	DroughtTolerance	Drought Response	Enhanced water efficiency

1NF Compliance: - ✓ **Atomic Values:** Single gene_code, category, trait_name per trait (no nested or composite structures) - ✓ **No Repeating Groups:** One category per genetic trait (not multiple categories per trait) - ✓ **Unique Identification:** trait_id uniquely identifies each genetic trait record

Table 4: Soil_Conditions

soil_id	region	soil_type	ph_level	nitrogen_content	phosphorus_content	observation_date
1	North Plain	Loam	7.2	45.5	32.1	2024-03-15
2	Valley Region	Clay	6.8	52.3	28.5	2024-03-20
3	Mountain Zone	Sandy Loam	7.5	38.2	24.7	2024-03-25

1NF Compliance: - ✓ **Atomic Values:** All numeric values (ph_level, nitrogen_content, phosphorus_content) are single atomic values, not composite measurements - ✓ **No Repeating Groups:** One soil_type and one set of measurements per observation record - ✓ **Unique Identification:** soil_id uniquely identifies each soil condition record

Table 5: Climate_Data

climate_id	region	temperature	humidity	rainfall	observation_date
1	North Plain	22.5	65.3	125.4	2024-03-15
2	Valley Region	24.1	72.1	95.2	2024-03-20
3	Mountain Zone	18.3	58.9	156.8	2024-03-25

1NF Compliance: - ✓ **Atomic Values:** Single numeric values for temperature, humidity, and rainfall (no composite climate readings) - ✓ **No Repeating Groups:** One set of climate measurements per observation (not multiple readings like temp1, temp2) - ✓ **Unique Identification:** climate_id uniquely identifies each climate observation record

Table 6: Performance_Indicators

performance_id	crop_id	yield_value	growth_rate	disease_resistance	water_efficiency	observation_date
1	1	5200.50	3.2	High	2.45	2024-03-15
2	2	4800.75	2.8	Very High	2.18	2024-03-20
3	3	6100.25	3.5	Medium	2.78	2024-03-25

1NF Compliance: - ✓ **Atomic Values:** Single numeric values for yield_value, growth_rate, water_efficiency (not composite indicators) - ✓ **No Repeating Groups:** One set of performance metrics per observation (not repeating metric sets) - ✓ **Unique Identification:** performance_id uniquely identifies each performance record

Table 7: Research_Analysis

analysis_id	trait_id	crop_id	performance_id	correlation_score	method	remarks
1	1	1	1	0.7850	Pearson	Strong positive correlation with yield
2	2	2	2	0.6240	Pearson	Moderate correlation with disease resistance
3	3	3	3	0.8120	Spearman	Strong correlation with water efficiency

1NF Compliance: - ✓ **Atomic Values:** Single correlation_score and method per analysis (not composite or multi-valued) - ✓ **No Repeating Groups:** One analysis method and

result per record (not multiple methods in one cell) - ✓ **Unique Identification:** analysis_id uniquely identifies each analysis record

Table 8: Audit_Logs

log_id	user_id	table_name	operation	record_id	action_time	details
1	2	Crop_Info	INSERT	5	2024-04-10 14:25	New crop variety added
2	2	Performance_Indicators	UPDATE	3	2024-04-10 15:10	Updated yield value
3	1	Genetic_Traits	DELETE	7	2024-04-10 16:45	Removed outdated trait

1NF Compliance: - ✓ **Atomic Values:** Single operation type, timestamp, and details per log entry (not nested or composite) - ✓ **No Repeating Groups:** One operation per audit record (not multiple operations in one log entry) - ✓ **Unique Identification:** log_id uniquely identifies each audit log entry

Table 9: Parents_References

reference_id	trait_id	crop_id	description	storage_type
1	1	1	Parent crop reference for Wheat	document
2	2	2	Parent crop reference for Rice	sequence file
3	3	3	Parent crop reference for Corn	document

1NF Compliance: - ✓ **Atomic Values:** Single description and storage_type per reference (not composite storage locations) - ✓ **No Repeating Groups:** One storage type per parent reference (not multiple storage types like storage1, storage2) - ✓ **Unique Identification:** reference_id uniquely identifies each parent reference record

SECOND NORMALIZATION TABLES (2NF)

Overview

2NF requires that all tables be in 1NF and have **no partial dependencies** - meaning all non-key attributes must depend on the entire primary key, not just part of it.

1. Users (2NF)

Short Description: Master user management table containing user authentication and role information.

Primary Key: user_id (single column)

Reasoning - Why Same as 1NF: - All attributes (username, email, role, created_at) fully depend on the entire primary key (user_id) - No partial dependencies exist - no non-key attribute depends on only part of the primary key - All attributes describe the user entity completely - **No structural changes needed from 1NF → 2NF**

2. Crop_Info (2NF)

Short Description: Master table containing crop information and characteristics.

Primary Key: crop_id (single column)

Reasoning - Why Same as 1NF: - All attributes (crop_name, variety, origin_region, description) fully depend on the entire primary key (crop_id) - No partial dependencies exist - no non-key attribute depends on only part of the primary key - All attributes are directly related to the crop entity - **No structural changes needed from 1NF → 2NF**

3. Genetic_Traits (2NF)

Short Description: Contains genetic trait information linked to crop types through foreign key relationships.

Primary Key: trait_id (single column)

Reasoning - Why Same as 1NF: - All non-key attributes (crop_id, gene_code, category, trait_name, description) fully depend on trait_id - No attribute depends on only part of the primary key - crop_id as foreign key does not create partial dependency - All attributes describe the genetic trait completely - **No structural changes needed from 1NF → 2NF**

4. Soil_Conditions (2NF)

Short Description: Environmental soil data collected from different regions with chemical analysis.

Primary Key: soil_id (single column)

Reasoning - Why Same as 1NF: - All non-key attributes (region, soil_type, ph_level, nitrogen_content, phosphorus_content, observation_date) fully depend on soil_id - No attribute depends on a subset of the primary key - All chemical properties are independent measurements - Region and soil_type both describe the soil condition, not just part of it -

No structural changes needed from 1NF → 2NF

5. Climate_Data (2NF)

Short Description: Meteorological observations recorded for different regions at specific dates.

Primary Key: climate_id (single column)

Reasoning - Why Same as 1NF: - All attributes (region, temperature, humidity, rainfall, observation_date) depend entirely on climate_id - No partial dependency - each non-key attribute is fully functional dependent on the primary key - Temperature, humidity, and rainfall are independent measurements, not derived from each other - Region identifies the location but still depends on climate_id -

No structural changes needed from 1NF → 2NF

6. Performance_Indicators (2NF)

Short Description: Crop performance metrics measured for different varieties over time.

Primary Key: performance_id (single column)

Reasoning - Why Same as 1NF: - All non-key attributes (crop_id, yield_value, growth_rate, disease_resistance, water_efficiency, observation_date) fully depend on performance_id - No attribute depends on only part of the primary key - crop_id as a foreign key does not violate 2NF - All performance metrics are independent measurements for the same observation -

No structural changes needed from 1NF → 2NF

7. Research_Analysis (2NF)

Short Description: Correlation analysis results between genetic traits and crop performance metrics.

Primary Key: analysis_id (single column)

Reasoning - Why Same as 1NF: - All non-key attributes (trait_id, crop_id, performance_id, correlation_score, method, remarks) fully depend on analysis_id - No partial dependency - all attributes are associated with the analysis record - Foreign keys (trait_id, crop_id, performance_id) do not cause partial dependencies - Correlation score, method, and remarks all describe one analysis, not part of it - **No structural changes needed from 1NF → 2NF**

8. Audit_Logs (2NF)

Short Description: System audit trail tracking all database operations with user and timestamp information.

Primary Key: log_id (single column)

Reasoning - Why Same as 1NF: - All non-key attributes (user_id, table_name, operation, record_id, action_time, details) fully depend on log_id - No attribute depends on only part of the primary key - user_id as foreign key does not create partial dependency - All attributes describe the audit event completely - **No structural changes needed from 1NF → 2NF**

9. Parents_References (2NF)

Short Description: Reference data for parent crop varieties and genetic material storage information.

Primary Key: reference_id (single column)

Reasoning - Why Same as 1NF: - All non-key attributes (trait_id, crop_id, description, storage_type) fully depend on reference_id - No attribute depends on only part of the primary key - trait_id and crop_id as foreign keys do not create partial dependencies - All attributes describe the parent reference completely - **No structural changes needed from 1NF → 2NF**

Key Insight for 2NF:

All 9 tables use single-column primary keys. Since partial dependencies can only occur with composite (multi-column) primary keys, and each table has only one primary key column, **all non-key attributes automatically depend on the entire primary key**. Therefore, all tables remain structurally identical from 1NF to 2NF.

THIRD NORMALIZATION TABLES (3NF)

Overview

3NF requires that all tables be in 2NF and have **no transitive dependencies** - meaning no non-key attribute should depend on another non-key attribute. All attributes must directly describe only the primary key entity.

3NF Entity Schema

Entity Name	Primary Key	Foreign Keys	Description
Users	user_id	None	User authentication and role management; no transitive dependencies
Crop_Info	crop_id	None	Master crop data - no dependencies on other entities
Genetic_Traits	trait_id	crop_id	Trait information; crop_id references Crop_Info
Soil_Conditions	soil_id	None	Soil measurements - no transitive dependencies; region is independent
Climate_Data	climate_id	None	Climate measurements - no transitive dependencies; region is independent
Performance_Indicators	performance_id	crop_id	Crop performance metrics; crop_id references Crop_Info
Research_Analysis	analysis_id	trait_id, crop_id, performance_id	Analysis results; references Genetic_Traits, Crop_Info, Performance_Indicators
Audit_Logs	log_id	user_id	Audit trail; user_id references Users

Entity Name	Primary Key	Foreign Keys	Description
Parents_References	reference_id	trait_id, crop_id	Parent references; trait_id references Genetic_Traits, crop_id references Crop_Info

Transitive Dependency Analysis:

Users: - No transitive dependencies - All attributes describe the user directly - Status: ✓
3NF Compliant

Crop_Info: - No transitive dependencies - All attributes describe the crop directly - Status:
✓ 3NF Compliant

Genetic_Traits: - No transitive dependencies - gene_code does not determine category -
trait_name does not determine description - Status: ✓ 3NF Compliant

Soil_Conditions: - No transitive dependencies - region does not determine soil_type
(independent measurements) - soil_type does not determine ph_level (independent
properties) - ph_level does not determine nitrogen_content (independent values) - Status:
✓ 3NF Compliant

Climate_Data: - No transitive dependencies - region does not determine temperature -
temperature does not determine humidity - humidity does not determine rainfall (all
independent measurements) - Status: ✓ 3NF Compliant

Performance_Indicators: - No transitive dependencies - yield_value does not determine
growth_rate - growth_rate does not determine disease_resistance - All metrics are
independent measurements - Status: ✓ 3NF Compliant

Research_Analysis: - No transitive dependencies - correlation_score does not determine
method - method does not determine remarks - All attributes directly describe the analysis
record - Status: ✓ 3NF Compliant

Audit_Logs: - No transitive dependencies - operation does not determine action_time -
table_name does not determine operation - All attributes directly describe the audit event -
Status: ✓ 3NF Compliant

Parents_References: - No transitive dependencies - description does not determine
storage_type - storage_type does not determine any other attribute - All attributes directly
describe the parent reference - Status: ✓ 3NF Compliant

Key Insight for 3NF:

All 9 tables satisfy 3NF requirements because: 1. No non-key attribute depends on another non-key attribute 2. All attributes directly describe the primary key entity 3. No attribute chains exist ($A \rightarrow B \rightarrow C$ dependencies) 4. Foreign keys reference other entities, not creating transitive dependencies within the table